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November 13, 2026

The Editor
Probability in the Engineering and Informational Sciences

Dear Editor,

I submit the manuscript "Two end modes: logarithmic asymptotics and importance sampling for the peak load of a pickup-and-delivery route" for consideration in *Probability in the Engineering and Informational Sciences*.

The paper studies the tail of the peak of a load process that arises in vehicle routing: a maximum of sums of gamma variables coupled by one-factor Gaussian copulas. It proves logarithmic asymptotics with explicit spectral scales, a sharp threshold for the efficiency of single-shift importance sampling, an exact description of how the single shift underestimates without inflating its own reported variance, and a sufficient condition, reduced to a computable covering constant, for the logarithmic efficiency of a mixture of shifts. A short numerical section checks the conditions and predictions on routes of two regional plans.

In the interest of transparency: an earlier, computation-oriented version of this work was submitted to the *INFORMS Journal on Computing* (JOC-2026-10-OA-2667). The area editor, Prof. Shane Henderson, declined it without review because the model is narrow, noted that the analytical treatment of the importance-sampling problem was its valuable part, and suggested *Probability in the Engineering and Informational Sciences* as an outlet. The present manuscript has been rewritten around that analysis: all proofs are complete (the earlier version gave sketches), the results on the silent underestimation and on mixture efficiency are new, and the computational material has been reduced to an illustration.

The manuscript is not under consideration elsewhere. Code and data are publicly available. There are no competing interests and no funding to declare.

Sincerely,

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